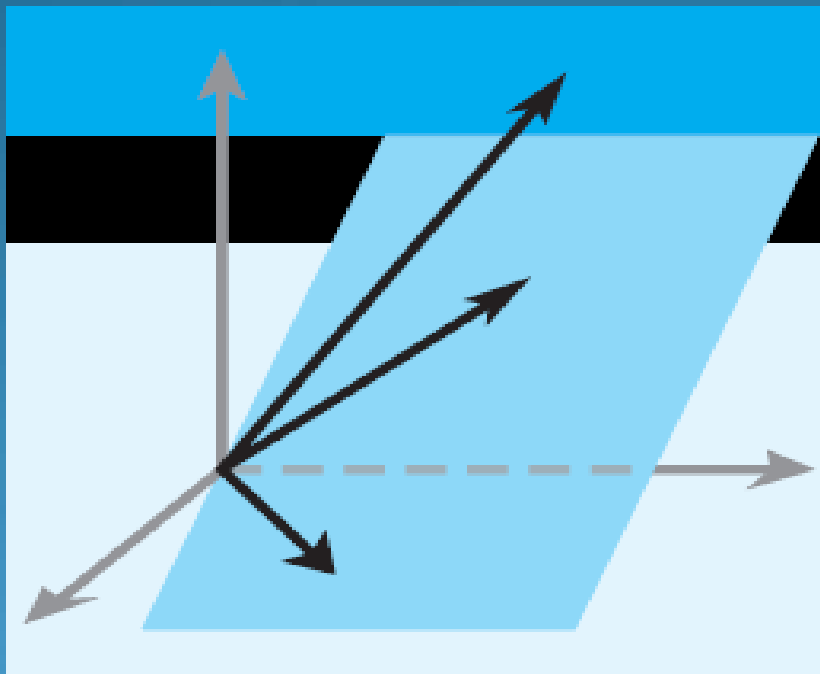


Elementary Linear Algebra



Chapter 4

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Chapter 4 General Vector Spaces

- 4.1 Real Vector Spaces
- 4.2 Subspaces
- 4.3 Linear Independence
- 4.4 Coordinates and Basis
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Section 4.6 Change of Basis

► EXAMPLE 1 Finding Transition Matrices

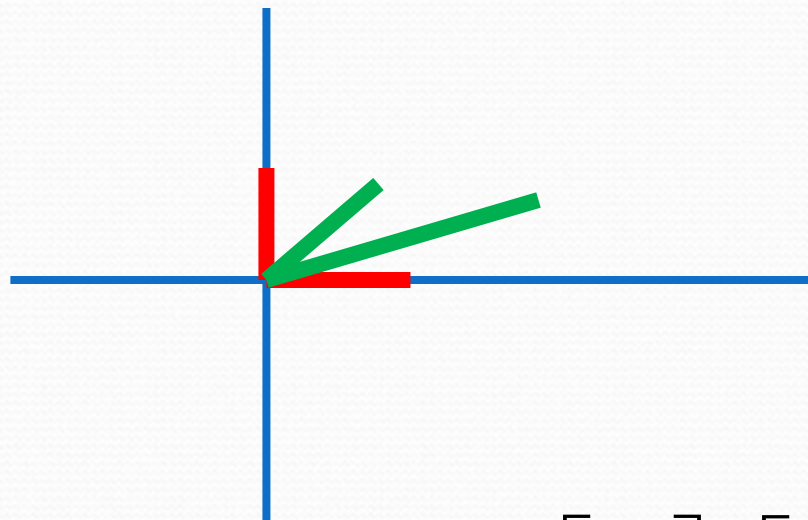
Consider the bases $B = \{ \mathbf{u}_1, \mathbf{u}_2 \}$ and $B' = \{ \mathbf{u}'_1, \mathbf{u}'_2 \}$ for R^2 , where

$$\mathbf{u}_1 = (1, 0), \quad \mathbf{u}_2 = (0, 1), \quad \mathbf{u}'_1 = (1, 1), \quad \mathbf{u}'_2 = (2, 1)$$

- (a) Find the transition matrix $P_{B' \rightarrow B}$ from B' to B .
- (b) Find the transition matrix $P_{B \rightarrow B'}$ from B to B' .

Section 4.6 Change of Basis

The Change-of-Basis Problem If $\mathbf{v}$ is a vector in a finite-dimensional vector space V , and if we change the basis for V from a basis B to a basis B' , how are the coordinate vectors $[\mathbf{v}]_B$ and $[\mathbf{v}]_{B'}$ related?



$$\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 2 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

Section 4.6 Change of Basis

$$B = \{u_1, u_2\}$$

$$B' = \{u'_1, u'_2\}$$


$$\begin{cases} u'_1 = au_1 + bu_2 \\ u'_2 = cu_1 + du_2 \end{cases}$$

$$v \in V$$


$$v = k_1 u'_1 + k_2 u'_2 \quad \longrightarrow \quad v = k_1 (au_1 + bu_2) + k_2 (cu_1 + du_2)$$

$$[v]_{B'} = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \quad \longrightarrow \quad [v]_B = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

Section 4.6 Change of Basis

$$B = \{u_1, u_2\}$$

$$B' = \{u'_1, u'_2\}$$


$$\begin{cases} u'_1 = au_1 + bu_2 \\ u'_2 = cu_1 + du_2 \end{cases}$$


$$v = k_1u'_1 + k_2u'_2 \quad \longrightarrow \quad v = k_1(au_1 + bu_2) + k_2(cu_1 + du_2)$$

$$\longrightarrow \quad v = (ak_1 + ck_2)u_1 + (bk_1 + dk_2)u_2$$

$$\longrightarrow \quad v = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

$$[v]_{B'} = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \quad \longrightarrow \quad [v]_B = \begin{bmatrix} ? \\ ? \end{bmatrix} = \begin{bmatrix} ak_1 + ck_2 \\ bk_1 + dk_2 \end{bmatrix}$$

Section 4.6 Change of Basis

Solution of the Change-of-Basis Problem If we change the basis for a vector space V from an old basis $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ to a new basis $B' = \{\mathbf{u}'_1, \mathbf{u}'_2, \dots, \mathbf{u}'_n\}$, then for each vector $\mathbf{v}$ in V , the old coordinate vector $[\mathbf{v}]_B$ is related to the new coordinate vector $[\mathbf{v}]_{B'}$ by the equation

$$[\mathbf{v}]_B = P[\mathbf{v}]_{B'} \quad (7)$$

where the columns of P are the coordinate vectors of the new basis vectors relative to the old basis; that is, the column vectors of P are

$$[\mathbf{u}'_1]_B, \quad [\mathbf{u}'_2]_B, \dots, \quad [\mathbf{u}'_n]_B \quad (8)$$

Section 4.6 Change of Basis

Solution (a) Here the old basis vectors are $\mathbf{u}'_1$ and $\mathbf{u}'_2$ and the new basis vectors are $\mathbf{u}_1$ and $\mathbf{u}_2$. We want to find the coordinate matrices of the old basis vectors $\mathbf{u}'_1$ and $\mathbf{u}'_2$ relative to the new basis vectors $\mathbf{u}_1$ and $\mathbf{u}_2$. To do this, observe that

$$\begin{aligned}\mathbf{u}'_1 &= \mathbf{u}_1 + \mathbf{u}_2 \\ \mathbf{u}'_2 &= 2\mathbf{u}_1 + \mathbf{u}_2\end{aligned}$$

from which it follows that

$$[\mathbf{u}'_1]_B = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad [\mathbf{u}'_2]_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and hence that

$$P_{B' \rightarrow B} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

Section 4.6 Change of Basis

Solution (b) Here the old basis vectors are $\mathbf{u}_1$ and $\mathbf{u}_2$ and the new basis vectors are $\mathbf{u}'_1$ and $\mathbf{u}'_2$. As in part (a), we want to find the coordinate matrices of the old basis vectors $\mathbf{u}'_1$ and $\mathbf{u}'_2$ relative to the new basis vectors $\mathbf{u}_1$ and $\mathbf{u}_2$. To do this, observe that

$$\begin{aligned}\mathbf{u}_1 &= -\mathbf{u}'_1 + \mathbf{u}'_2 \\ \mathbf{u}_2 &= 2\mathbf{u}'_1 - \mathbf{u}'_2\end{aligned}$$

from which it follows that

$$[\mathbf{u}_1]_{B'} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \text{and} \quad [\mathbf{u}_2]_{B'} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

and hence that

$$P_{B \rightarrow B'} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$



Section 4.6 Change of Basis

► EXAMPLE 2 Computing Coordinate Vectors

Let B and B' be the bases in Example 1. Use an appropriate formula to find $[\mathbf{v}]_B$ given that

$$[\mathbf{v}]_{B'} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

Solution

To find $[\mathbf{v}]_B$ we need to make the transition from B' to B . It follows from Formula 11 and part (a) of Example 1 that

$$[\mathbf{v}]_B = P_{B' \rightarrow B} [\mathbf{v}]_{B'} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

Section 4.6 Change of Basis

THEOREM 4.6.1 *If P is the transition matrix from a basis B' to a basis B for a finite-dimensional vector space V , then P is invertible and P^{-1} is the transition matrix from B to B' .*



Section 4.6 Change of Basis

▶ EXAMPLE 3 Example [1](#) Revisited

In Example [1](#) we considered the bases $B = \{ \mathbf{u}_1, \mathbf{u}_2 \}$ and $B' = \{ \mathbf{u}'_1, \mathbf{u}'_2 \}$ for R^2 , where

$$\mathbf{u}_1 = (1, 0), \quad \mathbf{u}_2 = (0, 1), \quad \mathbf{u}'_1 = (1, 1), \quad \mathbf{u}'_2 = (2, 1)$$

- (a) Use Formula [14](#) to find the transition matrix from B' to B .
- (b) Use Formula [14](#) to find the transition matrix from B to B' .

Section 4.6 Change of Basis

Solution (a) Here B' is the old basis and B is the new basis, so

$$[\text{new basis} \mid \text{old basis}] = \left[\begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

Since the left side is already the identity matrix, no reduction is needed. We see by inspection that the transition matrix is

$$P_{B' \rightarrow B} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

which agrees with the result in Example [1](#).

Section 4.6 Change of Basis

Solution (b) Here B is the old basis and B' is the new basis, so

$$[\text{new basis} \mid \text{old basis}] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right]$$

By reducing this matrix, so the left side becomes the identity, we obtain (verify)

$$[I \mid \text{transition from old to new}] = \left[\begin{array}{cc|cc} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \end{array} \right]$$

so the transition matrix is

$$P_{B \rightarrow B'} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

which also agrees with the result in Example [1](#).



Section 4.6 Change of Basis

THEOREM 4.6.2

Let $B' = \{ \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \}$ be any basis for the vector space $\mathbb{R}^n$ and let $S = \{ \mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \}$ be the standard basis for $\mathbb{R}^n$. If the vectors in these bases are written in column form, then

$$P_{B' \rightarrow S} = [\mathbf{u}_1 | \mathbf{u}_2 | \dots | \mathbf{u}_n] \quad (15)$$

Homework

Sec. 4.6

True/False Page 214

105 (a) (b), pick one Page 277

106 (a) (b), pick one

107

108

111

125

This is it

